AI 500 - Foundations of AI

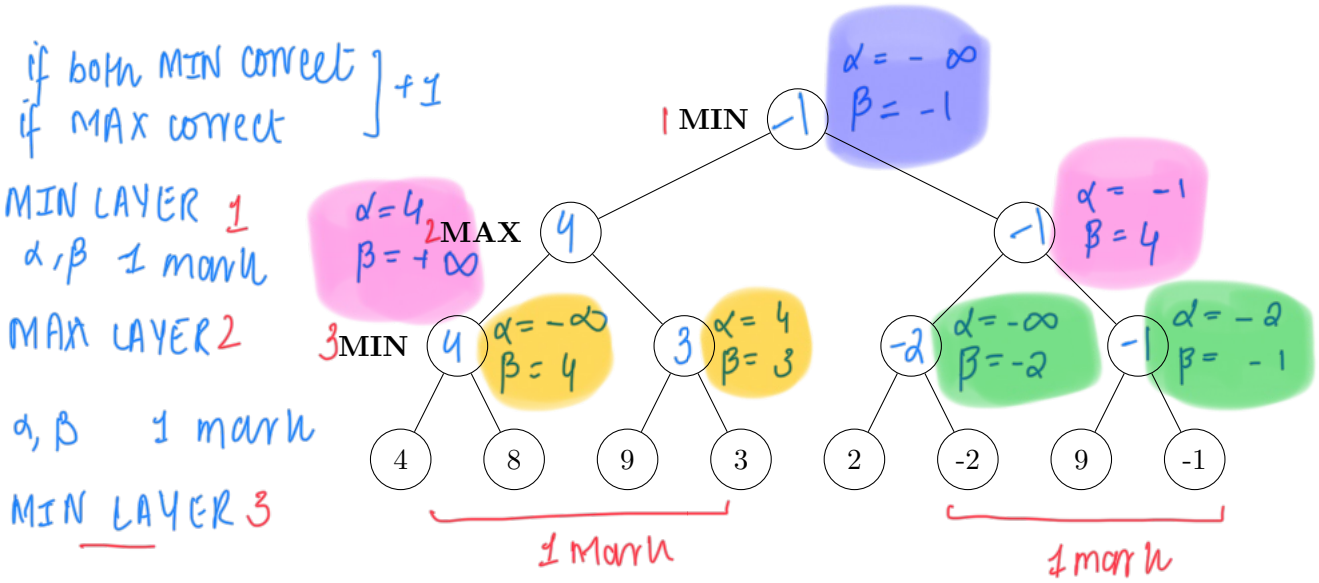
QUIZ 2

September 27, 2025

Name: _____ **Roll Number:** _____

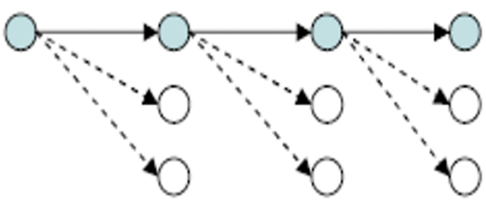
Q1) Alpha-Beta Pruning [5 marks]

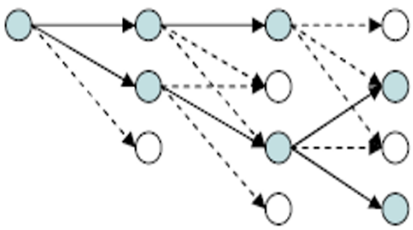
Given the search tree below, find the **MIN value** for the topmost node, and state which is the **optimal path** to follow for this node (left or right). Also, since you are applying minimax with **alpha-beta pruning**, prune the relevant branches (you may do so by slashing the relevant branch).



Q2) Beam Search vs Greedy Search [5 marks]

Consider the two search graphs shown below. Each node represents a partial state in the search process, with edges indicating possible expansions. One graph corresponds to **Greedy Search** while the other corresponds to **Beam Search**. Identify which graph represents **Greedy Search** and which represents **Beam Search**. Justify your answer by explaining the differences in how nodes are expanded in each search method.





Left Graph represents: Greedy Search (1)

Right Graph represents: Beam Search (1)

Beam width (for Beam Search): 2 (1)

Reasoning: (2)
Graph 1 only single node based on heuristic value is expanded! single path is being followed and all other are being ignored.
Graph 2 explores multiple tracks based on beam width value.

Q3) Recursive Best-First Search (RBFS) **[2 marks]**

Which of the following statements accurately describe the behavior of the Recursive Best-First Search (RBFS) algorithm?

A. To save memory, RBFS maintains a strict "closed list" and never re-expands a node that has been visited before.

B. The algorithm's decision to continue exploring a path or to backtrack is guided by an f-limit, which represents the f-cost of the best alternative path available from an ancestor.

C. RBFS is generally faster than A* search because its recursive nature explores fewer nodes overall.

D. When the algorithm backtracks from a node, it updates that node's stored f-value to reflect the best f-value of the promising descendant path that was just abandoned.

E. The algorithm primarily relies on the heuristic cost $h(n)$ to make decisions, ignoring the actual path cost $g(n)$ when comparing nodes.